



DIGITAL ASSIGNMENT 2

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Course: Big Data Analytics Lab

Course Code: PMDS507P

Start:

- ➔ start-dfs.sh
- ➔ start-yarn.sh

```
hduser@sjt217score047:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [sjt217score047]
2025-09-12 12:06:11,432 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
hduser@sjt217score047:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
hduser@sjt217score047:~$
```

Hadoop DFS running on localhost:9870

Check Process Status

➔ jps

```
hduser@sjt217score047:~$ jps
11120 Jps
9392 NameNode
10323 ResourceManager
9924 SecondaryNameNode
9599 DataNode
10479 NodeManager
hduser@sjt217score047:~$
```

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
dwxr-xr-x	hduser	supergroup	0 B	Mar 26 13:28	0	0 B	d_dir
dwxr-xr-x	hduser	supergroup	0 B	Mar 26 13:03	0	0 B	h_dir
dwxr-xr-x	hduser	supergroup	0 B	Mar 26 13:27	0	0 B	s_dir

List Root Directories:

➔ `hdfs dfs-ls /`

```
hduser@sjt217score047:~$ hdfs dfs -ls /
2025-09-12 12:23:56,990 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 3 items
drwxr-xr-x - hduser supergroup          0 2025-03-26 13:28 /d_dir
drwxr-xr-x - hduser supergroup          0 2025-03-26 13:03 /h_dir
drwxr-xr-x - hduser supergroup          0 2025-03-26 13:27 /s_dir
hduser@sjt217score047:~$
```

The screenshot shows the Hadoop web interface at localhost:9870/explorer.html. The 'Browse Directory' section is active, showing the root path '/'. A table lists three directories: /d_dir, /h_dir, and /s_dir. Each entry shows its permissions (drwxr-xr-x), owner (hduser), group (supergroup), size (0 B), last modified date (Mar 26 13:28, 13:03, and 13:27 respectively), replication (0), and block size (0 B). The interface includes a search bar, a 'Go!' button, and pagination controls showing 'Showing 1 to 3 of 3 entries'.

Create Directory:

➔ `hdfs dfs-mkdir /soumyadeep` (filename -> soumyadeep)

```
hduser@sjt217score047:~$ hdfs dfs -mkdir /soumyadeep
2025-09-12 12:41:56,880 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
hduser@sjt217score047:~$ hdfs dfs -ls /
2025-09-12 12:42:17,422 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 4 items
drwxr-xr-x - hduser supergroup          0 2025-03-26 13:28 /d_dir
drwxr-xr-x - hduser supergroup          0 2025-03-26 13:03 /h_dir
drwxr-xr-x - hduser supergroup          0 2025-03-26 13:27 /s_dir
drwxr-xr-x - hduser supergroup          0 2025-09-12 12:41 /soumyadeep
hduser@sjt217score047:~$
```

Create Multiple Directories at once:

➔ `hdfs dfs-mkdir-p /soumyadeep/dir1`

Put (upload) a file to HDFS :

➔ `hdfs dfs-put Desktop/soumyadeep.txt /Soumyadeep`

```
hduser@sjt217score047:~$ hdfs dfs -put Desktop/soumyadeep.txt /soumyadeep
2025-09-12 12:52:41,566 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
hduser@sjt217score047:~$
```

The screenshot shows the Hadoop web interface at `localhost:9870/explorer.html#/soumyadeep`. The 'Browse Directory' section shows a table with one entry:

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	hduser	supergroup	56 B	Sep 12 12:52	1	128 MB	soumyadeep.txt

Showing 1 to 1 of 1 entries

Get (download) a file from HDFS:

➔ `hdfs dfs-get /soumyadeep/file2.txt`

```
hduser@sjt217score047:~$ hdfs dfs -get /soumyadeep/file2.txt
2025-09-12 13:00:57,152 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
hduser@sjt217score047:~$
```

Remove a file :

➔ `hdfs dfs-rm /myfolder/data.txt`

```
hduser@sjt217score047:~$ hdfs dfs -rm /soumyadeep/file2.txt
2025-09-12 13:02:04,042 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Deleted /soumyadeep/file2.txt
hduser@sjt217score047:~$
```

The screenshot shows the Hadoop web interface at `localhost:9870/explorer.html#/soumyadeep`. The 'Browse Directory' section shows a table with one entry:

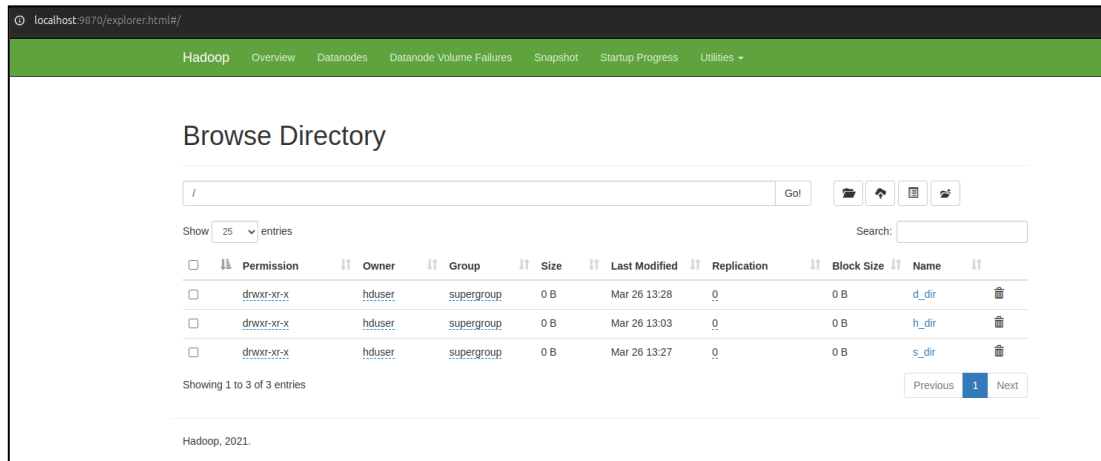
Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	hduser	supergroup	56 B	Sep 12 12:52	1	128 MB	soumyadeep.txt

Showing 1 to 1 of 1 entries

Remove a file Recursively: (This will delete files/folders permanently)

➔ `hdfs dfs-rm -r /Soumyadeep`

```
this is the 1st file in hadoop. We are learning hadoop.
hduser@sjt217score047:~$ hdfs dfs -rm -r /soumyadeep
2025-09-12 13:06:01,723 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform
... using builtin-java classes where applicable
Deleted /soumyadeep
hduser@sjt217score047:~$
```



View file contents :

➔ `hdfs dfs-cat /soumyadeep/soumyadeep.txt`

```
hduser@sjt217score047:~$ hdfs dfs -cat /soumyadeep/soumyadeep.txt
2025-09-12 13:03:58,139 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform
... using builtin-java classes where applicable
this is the 1st file in hadoop. We are learning hadoop.
hduser@sjt217score047:~$
```

Display first few lines:

➔ `hdfs dfs-head /soumyadeep /soumyadeep.txt`

Display last few lines:

➔ `hdfs dfs-tail / soumyadeep / soumyadeep.txt`

```
hduser@sjt217score047:~$ hdfs dfs -head /soumyadeep/soumyadeep.txt
2025-09-12 13:04:44,362 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform
... using builtin-java classes where applicable
this is the 1st file in hadoop. We are learning hadoop.
hduser@sjt217score047:~$ hdfs dfs -tail /soumyadeep/soumyadeep.txt
2025-09-12 13:04:57,420 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform
... using builtin-java classes where applicable
this is the 1st file in hadoop. We are learning hadoop.
hduser@sjt217score047:~$
```